

Derivation of Faraday and Kerr Rotation Angles

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March 10, 2025

1 Boundary Conditions

We have the conservation of charge on the interfaces

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{j} \quad (1)$$

and the Ohm's law

$$\mathbf{j} = \sigma \mathbf{E} = \begin{pmatrix} 0 & \sigma_{xy} \\ -\sigma_{xy} & 0 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} \sigma_{xy} E_y \\ -\sigma_{xy} E_x \end{pmatrix} \quad (2)$$

thus

$$\frac{\partial \rho}{\partial t} = -\sigma_{xy} \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) = \sigma_{xy} \frac{\partial B_z}{\partial t} \Rightarrow \rho = \sigma_{xy} B_z \quad (3)$$

Plugging this into the Maxwell equations, we have

$$\nabla \cdot \mathbf{E} = \frac{\sigma_{xy}}{\varepsilon_0} B_z \quad (4)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (5)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (6)$$

$$\nabla \times \mathbf{H} = \mathbf{j} + \varepsilon \frac{\partial \mathbf{E}}{\partial t} = \sigma \mathbf{E} \delta(z - z_0) + \varepsilon \frac{\partial \mathbf{E}}{\partial t} \quad (7)$$

We use the superscript R and L to distinguish right and left side of the interface and $+$ ($-$) to denote the direction of light propagation. By approximating the relative permeability constant in each region to 1, the electric and magnetic fields are given by (notice the wave vector is no longer perpendicular to the surface. But for simplicity, we just let the k_x of the incident light be 0, so that the k_x of both transmitted light and reflected light vanish)

$$\mathbf{B}^{+L} = \frac{n_L}{c} [(-E_y^{+L} \cos \theta + E_z^{+L} \sin \theta) \hat{x} + E_x^{+L} \cos \theta \hat{y} - E_x^{+L} \sin \theta \hat{z}] \quad (8)$$

$$\mathbf{B}^{-L} = \frac{n_L}{c} [(E_y^{-L} \cos \theta + E_z^{-L} \sin \theta) \hat{x} - E_x^{-L} \cos \theta \hat{y} - E_x^{-L} \sin \theta \hat{z}] \quad (9)$$

$$\mathbf{B}^{+R} = \frac{n_R}{c} [(-E_y^{+R} \cos \phi + E_z^{+R} \sin \phi) \hat{x} + E_x^{+R} \cos \phi \hat{y} - E_x^{+R} \sin \phi \hat{z}] \quad (10)$$

$$\mathbf{B}^{-R} = \frac{n_R}{c} [(E_y^{-R} \cos \phi + E_z^{-R} \sin \phi) \hat{x} - E_x^{-R} \cos \phi \hat{y} - E_x^{-R} \sin \phi \hat{z}] \quad (11)$$

where θ and ϕ represent the angle between the incident light, transmitted light and the interface normal, respectively, and they're connected through the law of refraction

$$n_L \sin \theta = n_R \sin \phi \quad (12)$$

Moreover, from the Maxwell equations we can derive the boundary conditions for $\mathbf{E}$ and $\mathbf{B}$

$$E_z^R - E_z^L = \frac{\sigma_{xy}}{\varepsilon_0} B_z^L \quad (13)$$

$$E_x^R = E_x^L \quad (14)$$

$$E_y^R = E_y^L \quad (15)$$

$$B_x^L - B_x^R = \mu_0 \sigma_{xy} E_x \quad (16)$$

$$B_y^L - B_y^R = \mu_0 \sigma_{xy} E_y \quad (17)$$

$$B_z^L = B_z^R \quad (18)$$

All components of electric field and magnetic field is unknown, which add up to 24 unknowns. To derive the transfer matrix, we set the incident components as known. Then the number of equations (18 equations) matches the unknowns.

2 Transfer Matrices

The boundary conditions obtained in section 1 are used in this section to derive the transfer matrix when light propagates through the interfaces. Since the z-component of electric field has a strict geometric constraint, or more precisely, uniquely determined by the incident angle and the y-component of electric field (two of our controllable variables). Moreover, through calculation, we find that the tangential components on the right side of the interface are uniquely determined by that on the left side. So if we just concern about the tangential components, the transfer matrix is still 4×4 .

$$\begin{pmatrix} E_x^{+R} \\ E_y^{+R} \\ E_x^{-R} \\ E_y^{-R} \end{pmatrix} = \mathbf{T}^I \begin{pmatrix} E_x^{+L} \\ E_y^{+L} \\ E_x^{-L} \\ E_y^{-L} \end{pmatrix} \quad (19)$$

where the transfer matrix $T^I(n_L, n_R, \sigma_{xy}, \theta)$ are displayed as

$$T^I = \frac{1}{2} \begin{pmatrix} 1 + \frac{n_L \cos \theta}{n_R \cos \phi} & -\frac{2\alpha\sigma_{xy}}{n_R \cos \phi} & 1 - \frac{n_L \cos \theta}{n_R \cos \phi} & -\frac{2\alpha\sigma_{xy}}{n_R \cos \phi} \\ -\frac{2\alpha\sigma_{xy}(n_L^2 \sin^2 \theta - 1)}{n_R \cos \phi} & 1 + \frac{n_L \cos \theta}{n_R \cos \phi} & -\frac{2\alpha\sigma_{xy}(n_L^2 \sin^2 \theta - 1)}{n_R \cos \phi} & 1 - \frac{n_L \cos \theta}{n_R \cos \phi} \\ 1 - \frac{n_L \cos \theta}{n_R \cos \phi} & \frac{2\alpha\sigma_{xy}}{n_R \cos \phi} & 1 + \frac{n_L \cos \theta}{n_R \cos \phi} & \frac{2\alpha\sigma_{xy}}{n_R \cos \phi} \\ \frac{2\alpha\sigma_{xy}(n_L^2 \sin^2 \theta - 1)}{n_R \cos \phi} & 1 - \frac{n_L \cos \theta}{n_R \cos \phi} & \frac{2\alpha\sigma_{xy}(n_L^2 \sin^2 \theta - 1)}{n_R \cos \phi} & 1 + \frac{n_L \cos \theta}{n_R \cos \phi} \end{pmatrix} \quad (20)$$

The Hall conductivity σ_{xy} is in a unit of $\frac{e^2}{h}$. Then, considering the interference of the light due to multiple reflections inside the film, the transfer matrix for the propagation in the topological insulator bulk can be regarded as an identity matrix since we're calculating all these things based on the long-wavelength approximation. The total transfer matrix is given by the product of the transfer matrices

$$T^{\text{total}} = T^I(n_f, n_s, \sigma_{xy}^b, \phi) T^I(n_v, n_f, \sigma_{xy}^t, \theta) \quad (21)$$

$$T^{\text{total}} = \frac{1}{2} \begin{pmatrix} 1 + \frac{n_v \cos \theta}{n_s \cos \gamma} & -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_s \cos \gamma} & 1 - \frac{n_v \cos \theta}{n_s \cos \gamma} & -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_s \cos \gamma} \\ -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} & 1 + \frac{n_v \cos \theta}{n_s \cos \gamma} & -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} & 1 - \frac{n_v \cos \theta}{n_s \cos \gamma} \\ 1 - \frac{n_v \cos \theta}{n_s \cos \gamma} & \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_s \cos \gamma} & 1 + \frac{n_v \cos \theta}{n_s \cos \gamma} & \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_s \cos \gamma} \\ \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} & 1 - \frac{n_v \cos \theta}{n_s \cos \gamma} & \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} & 1 + \frac{n_v \cos \theta}{n_s \cos \gamma} \end{pmatrix} \quad (22)$$

where the angles follow the law of refraction

$$n_v \sin \theta = n_f \sin \phi = n_s \sin \gamma \quad (23)$$

We can do a little estimation here. The thickness of the substrate is about 300 μm , but the duration of laser pulse is merely 100 fs. So the reflected wave from the back interface of the substrate is negligible when calculating the Faraday rotation angle. In this way, the transmitted and reflected light are related by the following equation

$$\begin{pmatrix} E_x^t \\ E_y^t \\ 0 \\ 0 \end{pmatrix} = T^{\text{total}} \begin{pmatrix} E_x^i \\ 0 \\ E_x^r \\ E_y^r \end{pmatrix} \quad (24)$$

That is

$$2E_x^t = \left(1 + \frac{n_v \cos \theta}{n_s \cos \gamma}\right) E_x^i + \left(1 - \frac{n_v \cos \theta}{n_s \cos \gamma}\right) E_x^r - \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_s \cos \gamma} E_y^r \quad (25)$$

$$2E_y^t = -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} E_x^i - \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} E_x^r + \left(1 - \frac{n_v \cos \theta}{n_s \cos \gamma}\right) E_y^r \quad (26)$$

$$0 = \left(1 - \frac{n_v \cos \theta}{n_s \cos \gamma}\right) E_x^i + \left(1 + \frac{n_v \cos \theta}{n_s \cos \gamma}\right) E_x^r + \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_s \cos \gamma} E_y^r \quad (27)$$

$$0 = \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} E_x^i + \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_v^2 \sin^2 \theta - 1)}{n_s \cos \gamma} E_x^r + \left(1 + \frac{n_v \cos \theta}{n_s \cos \gamma}\right) E_y^r \quad (28)$$

We get the Faraday rotation angle as

$$\tan \theta_F = \frac{E_y^t}{E_x^t} = \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(1 - n_v^2 \sin^2 \theta)}{n_s \cos \gamma + n_v \cos \theta} \quad (29)$$

The influence on Faraday rotation angle caused by the back interface is also negligible due to our previous estimation since we're focusing only on the amplitudes of E_y and E_x . The same applies to the Kerr rotation where the light is reflected at the substrate interface after experiencing the Faraday rotation. In this case, the total transfer matrix is rewritten as

$$\mathbf{T}^{\text{total}} = \mathbf{T}^{\text{I}}(n_f, n_v, \sigma_{xy}^t, \phi) \mathbf{T}^{\text{I}}(n_s, n_f, \sigma_{xy}^b, \gamma) \quad (30)$$

$$\mathbf{T}^{\text{total}} = \frac{1}{2} \begin{pmatrix} 1 + \frac{n_s \cos \gamma}{n_v \cos \theta} & -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_v \cos \theta} & 1 - \frac{n_s \cos \gamma}{n_v \cos \theta} & -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_v \cos \theta} \\ -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} & 1 + \frac{n_s \cos \gamma}{n_v \cos \theta} & -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} & 1 - \frac{n_s \cos \gamma}{n_v \cos \theta} \\ 1 - \frac{n_s \cos \gamma}{n_v \cos \theta} & \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_v \cos \theta} & 1 + \frac{n_s \cos \gamma}{n_v \cos \theta} & \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_v \cos \theta} \\ \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} & 1 - \frac{n_s \cos \gamma}{n_v \cos \theta} & \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} & 1 + \frac{n_s \cos \gamma}{n_v \cos \theta} \end{pmatrix} \quad (31)$$

In our measurement, the Kerr and Faraday rotation are combined together when we're obtaining the different components of the delayed second pulse. So here we shall still let θ to be the only variable. The transmitted and reflected light are related by following four equations

$$2E_x^t = \left(1 + \frac{n_s \cos \gamma}{n_v \cos \theta}\right) E_x^i + \left(1 - \frac{n_s \cos \gamma}{n_v \cos \theta}\right) E_x^r - \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_v \cos \theta} E_y^r \quad (32)$$

$$2E_y^t = -\frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} E_x^i - \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} E_x^r + \left(1 - \frac{n_s \cos \gamma}{n_v \cos \theta}\right) E_y^r \quad (33)$$

$$0 = \left(1 - \frac{n_s \cos \gamma}{n_v \cos \theta}\right) E_x^i + \left(1 + \frac{n_s \cos \gamma}{n_v \cos \theta}\right) E_x^r + \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)}{n_v \cos \theta} E_y^r \quad (34)$$

$$0 = \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} E_x^i + \frac{2\alpha(\sigma_{xy}^t + \sigma_{xy}^b)(n_s^2 \sin^2 \gamma - 1)}{n_v \cos \theta} E_x^r + \left(1 + \frac{n_s \cos \gamma}{n_v \cos \theta}\right) E_y^r \quad (35)$$

We get the Kerr rotation angle as

$$\tan \theta_K = \frac{4\alpha n_s(\sigma_{xy}^t + \sigma_{xy}^b)(1 - n_s^2 \sin^2 \gamma) \cos \gamma}{n_s^2 \cos^2 \gamma - n_v^2 \cos^2 \theta + 4\alpha^2(\sigma_{xy}^t + \sigma_{xy}^b)^2(n_s^2 \sin^2 \gamma - 1)} \quad (36)$$

or a more simplified form

$$\tan \theta_K = \frac{4\alpha n_s(\sigma_{xy}^t + \sigma_{xy}^b)(1 - n_v^2 \sin^2 \theta) \cos \gamma}{n_s^2 - n_v^2 + 4\alpha^2(\sigma_{xy}^t + \sigma_{xy}^b)^2(n_v^2 \sin^2 \theta - 1)} \quad (37)$$

When we're considering the z-component of the transmitted electric field, we find the refraction at the interface of the substrate and vacuum doesn't affect the ratio of x and y-component. In Faraday rotation, after reentering the vacuum, the relation between z and y-component is trigonometric as

$$E_z^t = -E_y^t \tan \theta \quad (38)$$

Using equation (29) and the relation between E_x^t and E_x^i , we have

$$\frac{E_z^t}{E_x^i} = -\frac{2\alpha n_v \sin 2\theta \cos \theta}{(n_s \cos \gamma + n_v \cos \theta)^2 - 4\alpha^2(\sigma_{xy}^t + \sigma_{xy}^b)^2(n_v^2 \sin^2 \theta - 1)} \quad (39)$$

For Streda charge and vacuum, we have

$$\frac{E_z^t}{E_x^i} = -\frac{2\alpha \sin 2\theta \cos \theta}{(\sqrt{n_s^2 - \sin^2 \theta} + \cos \theta)^2 + 4\alpha^2 \cos^2 \theta} \quad (40)$$

In Kerr rotation, after reentering the vacuum, we have

$$E_z^r = E_y^r \tan \theta \quad (41)$$

Using equation (36) and the relation between E_x^r and E_x^i , we have

$$\frac{E_z^r}{E_x^i} = \frac{4\alpha n_s(\sigma_{xy}^t + \sigma_{xy}^b)^2(1 - n_v^2 \sin^2 \theta) \tan \theta \cos \gamma}{(n_s \cos \gamma + n_v \cos \theta)^2 - 4\alpha^2(\sigma_{xy}^t + \sigma_{xy}^b)^2(n_v^2 \sin^2 \theta - 1)} \quad (42)$$

For Streda charge and vacuum, we have

$$\frac{E_z^r}{E_x^i} = \frac{2\alpha \sin 2\theta \cos \theta \sqrt{n_s^2 - \sin^2 \theta}}{(\sqrt{n_s^2 - \sin^2 \theta} + \cos \theta)^2 + 4\alpha^2 \cos^2 \theta} \quad (43)$$

3 Discussion

If we're talking about the Streda scenario, then both σ_{xy}^t and σ_{xy}^b are $\frac{1}{2}(\frac{e^2}{h})$, which means (refractive index for vacuum is 1)

$$\tan \theta_F = \frac{2\alpha \cos^2 \theta}{n_s \cos \gamma + \cos \theta} = \frac{2\alpha \cos^2 \theta}{\sqrt{n_s^2 - \sin^2 \theta} + \cos \theta} \quad (44)$$

$$\tan \theta_K = \frac{4\alpha n_s \cos^2 \theta \cos \gamma}{n_s^2 - 1 - 4\alpha^2 \cos^2 \theta} = \frac{4\alpha \cos^2 \theta \sqrt{n_s^2 - \sin^2 \theta}}{n_s^2 - 1 - 4\alpha^2 \cos^2 \theta} \quad (45)$$

Since the fine structure constant $\alpha \approx \frac{1}{137}$, so it's quite reasonable to validate the following approximation

$$\theta_F \approx \arctan \frac{2\alpha \cos^2 \theta}{\sqrt{n_s^2 - \sin^2 \theta} + \cos \theta} \quad (46)$$

$$\theta_K \approx \arctan \frac{4\alpha \cos^2 \theta \sqrt{n_s^2 - \sin^2 \theta}}{n_s^2 - 1} \quad (47)$$

While these two quantities depend on the refractive index of the substrate, we can extract α by combining them

$$\alpha = \frac{\tan \theta_F \tan \theta_K - \tan^2 \theta_F}{\tan \theta_K - 2 \tan \theta_F} \frac{1}{\cos \theta} \quad (48)$$

If we further assume the incident angle is small enough to validate the approximation $\sin \theta \approx \theta$, then we have

$$\theta_F = \arctan \frac{2\alpha}{n_s + 1} + \mathcal{O}(\alpha\theta^2) \quad (49)$$

$$\theta_K = \arctan \frac{4\alpha n_s}{n_s^2 - 1} + \mathcal{O}(\alpha\theta^2) \quad (50)$$

If this assumption is invalid, the following figure 1 might help. This figure precisely corresponds to Mogi's doctoral thesis when $\theta = 0$. Figure 2 is the visualized version of equation (40) and (43). Please click [Github repository](#) for the relevant code.

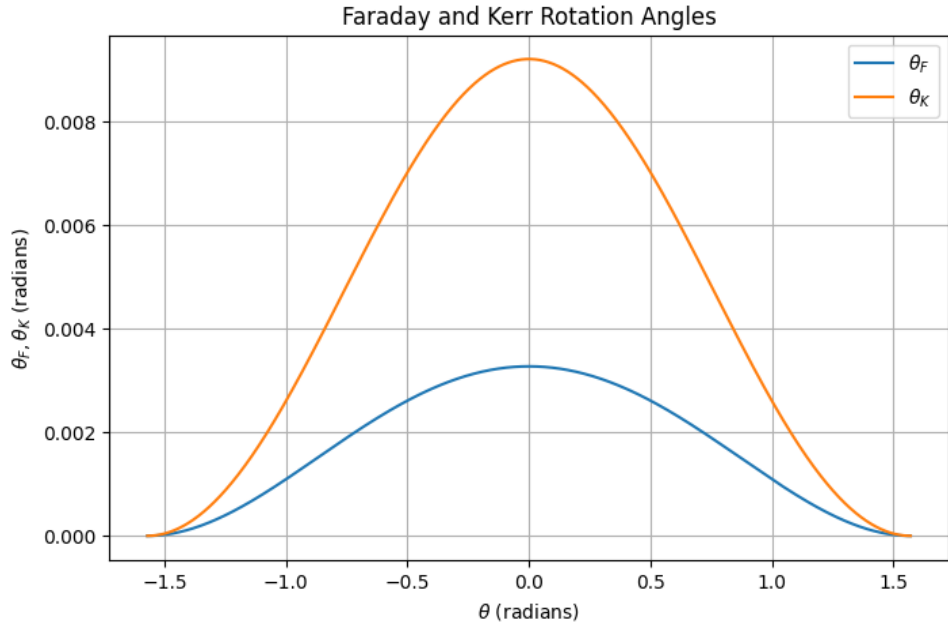


Figure 1: Relation Between Rotation Angles and Incident Angles

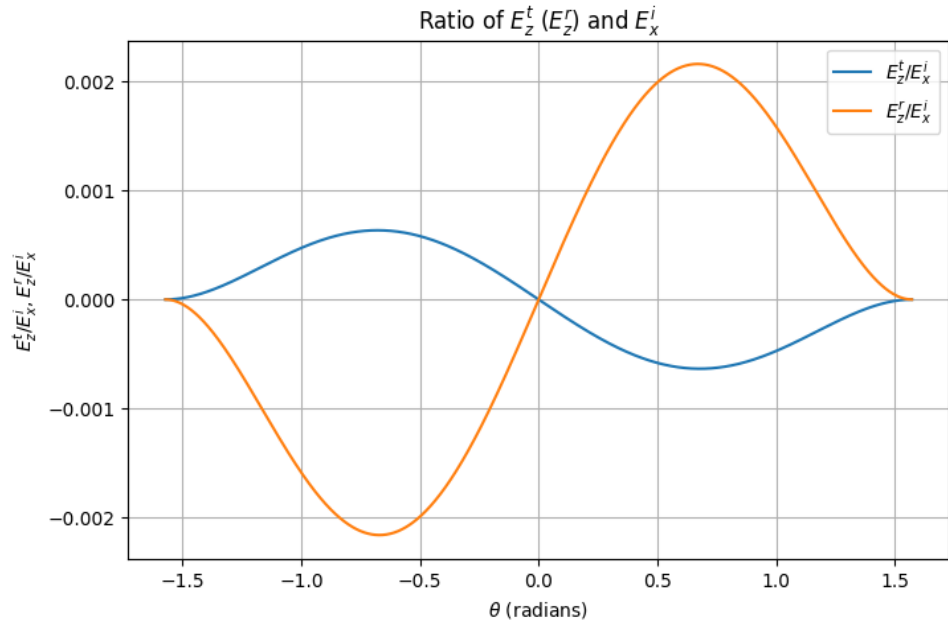


Figure 2: Ratio of $E_z^t(E_z^r)$ and E_x^i